PERSONAL / PUBLIC TECHNICAL PROJECT

ESP32 robotics *control platform.*

A hardware-backed robotics platform that brings sensing, operator input, motor drive and feedback control into one testable embedded system.

ESP32 / C++ IMU ENCODERS PID CONTROL

01 / OBJECTIVE

Close the loop *on hardware.*

The project integrates inertial sensing, wireless operator input, encoder feedback, motor drive, servo control and self-balancing PID behavior around an ESP32 platform.

> This is representative personal engineering work, not a completed client contract. Claims are limited to functions and technologies documented by the public repository.

02 / ARCHITECTURE

Signals into *behavior.*

Technical scope

Embedded C++ connects sensors, wireless commands, encoder feedback, PWM motor output and servo actuation. MATLAB/Simulink integration is documented as part of the project workflow.

INPUTS

IMU, wireless operator input and encoder feedback.

CONTROL

PID-based self-balancing behavior and actuator commands.

ARCHITECTURE PLACEHOLDER

SENSOR → CONTROLLER → ACTUATOR SIGNAL-FLOW DIAGRAM

Replace with an authorized architecture diagram.

Source: public repository documentation

PROTOTYPE PLACEHOLDER

AUTHORIZED HARDWARE PHOTOGRAPH

Prototype or bench image, when cleared for publication.

BEHAVIOR PLACEHOLDER

CONTROL RESPONSE PLOT

Measured response or software/hardware integration evidence.

03 / EVIDENCE

Public work with *clear limits.*

DOCUMENTED

Embedded integration

Firmware and platform documentation describe the sensing, actuation and feedback chain.

BOUNDARY

No invented metrics

No performance number, certification or production claim is made here.

STATUS

Public project

Repository available for technical review; source remains at the original public link.

VIEW REPOSITORY ↗

RELATED SERVICE: EMBEDDED CONTROLS →

START A TECHNICAL DISCUSSION

Bring the *technical question.*

Describe the system, its current stage and the evidence available. Secure file exchange can follow the initial discussion.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com